Isaac S. Narrett

https://isaacnarrett.github.io — narrett@mit.edu — MIT EAPS Ph.D. Candidate

Research Interests

My work focuses on magnetic fields, planetary evolution, and habitability, meeting at the intersection of planetary science and space physics. My research combines magnetic field modeling, magnetohydrodynamic (MHD) simulations, and spacecraft observations to investigate planetary processes and dive into the interiors of terrestrial bodies and subsurface oceans of icy moons.

Education

Massachusetts Institute of Technology (MIT) — Cambridge, MA, 2021-
Ph.D., Planetary Science

University of Michigan — Ann Arbor, MI, 2017-2021
B.S.E., Physics, *summa cum laude*
Minor, Space Science Engineering

Employment

Massachusetts Institute of Technology (MIT) — Cambridge, MA, 2021–
Graduate Student Researcher

Battelle Memorial Institute — Columbus, OH, Summer 2020
Engineering Modeler Intern

University of Michigan — Ann Arbor, MI, 2017-2021
Moldwin Magnetics Lab Research Assistant

Teaching and Mentorship

Undergrad research mentor, *magnetometer design*, MIT UROP	Spring 2026
Co-advisor for MIT AeroAstro Master's student, *rover magnetometer modeling*	Fall 2025-2026
Undergrad research mentor, *rover magnetometer modeling*, MIT UROP	Summer, Fall 2025
Summer mentor, *magnetometer data analysis*, Haverford College Student	Summer 2025
Graduate Teaching Assistant, Hands-on Astronomy	Spring 2024, 2025, 2026
Polygence Paid Mentor, King's College School, UK	Summer 2025
Mentor, Cambridge High School Student	Summer 2024

Professional Experience

Primary convener for AGU *Planetary Magnetism* session	Winter 2026
Executive secretary and external reviewer for NASA program review panel	Spring 2025
Convener for AGU *Planetary Magnetism* session	Winter 2025
Lunar Planetary Science Conference *Mercury from the Inside Out* moderator	Spring 2025
Reviewer for *Planetary Science Journal*, *Philosophical Transactions of the Royal Society A*	2024-

Honors, Grants, and Awards

MIT School of Science *MathWorks* Fellowship	2026
Co-I on James Webb Space Telescope (JWST) Cycle 4 DD Proposal	*Selected* 2025
Elected to Sigma Xi	2025

NASA FINESST Fellow Award	2023-2025
MIT - James L. Elliot Fellowship	2023
MIT - Presidential Fellowship	2021
Michigan - Al Levinson Memorial Scholarship	2020-2021
Michigan - Guezner Engineering Merit Scholarship	2019-2020

Service and Outreach

Proposal team, MIT QoL *Selected* Grant - K-12 Outreach Kit	Spring 2025
Visitor, Amigos School and Manning Elementary School	Spring 2024, 2025
Organizer, MIT EAPS Planetary Seminar	2023-
Organizer, Cambridge Science Carnival	2022-2025

Journal Articles

Published:

- **Narrett, I. S.**, Oran, R., Chen, Y., Miljkovic, K., Toth, G., Johnson, C. L., Weiss, B. P. (2026). Impact plasma amplification of the ancient Mercury magnetic field. *JGR Planets.*
- **Narrett, I. S.**, Oran, R., Chen, Y., Miljkovic, K., Toth, G., Mansbach, E. N., Weiss, B. P. (2025). Impact plasma amplification of the ancient lunar dynamo. *Science Advances.* Space.com Article
- **Narrett, I. S.**, Rackham, B. V., de Wit, J. (2024). Axisymmetric High Spot Coverage on Exoplanet Host HD 189733 A. *The Astronomical Journal.*
- de Wit, J., Rivkin, A. S., Micheli, M., Farnocchia, D., Burdanov, A.,..., **Narrett, I. S.**, et al. (2026). JWST Observations of Asteroid 2024 YR4 Rule Out a 2032 Lunar Impact and Demonstrate a New Regime for Planetary Defense Follow-up. *Astrophysical Journal Letters.*
- Strabel, B. P., Regoli, L. H., Moldwin, M. B., Ojeda, L. V., Shi, Y., Thomas, J. D., **Narrett, I. S.**, et al. (2022). Quad-Mag board for CubeSat applications. *Geoscientific Instrumentation, Methods and Data Systems.*

In Review/Preprint:

- **Narrett, I. S.**, Weiss, B. P., Steele, S. C., Biersteker, J. B. (2025). Mercury's Crustal Magnetization Requires a Stronger Ancient Dynamo. *In Review at PNAS. arXiv*

In Prep:

- **Narrett, I. S.**, Chen, Y., Biersteker, J. B., Jia, X., Weiss, B. P. A multi-fluid MHD model of the dust-plasma environment at Enceladus.
- Schneck, U. G., Biersteker, J. B., **Narrett, I. S.**, Jia, X., Weiss, B. P. Confirmation of an ocean on Callisto using Bayesian reanalysis of Galileo induction measurements.

Invited Talks

Magnetism of the Moon and Mercury, MIT Planetary Seminar	2026
Lunar and Hermean Magnetism, Boston University	2026
Lunar Impact-Plasma Dynamo Amplification and Future Tests, AGU Winter Conference	2025
Magnetizing the Moon and Mercury, UC Berkeley Planetary Science Seminar	2025
The Uranian Magnetosphere, Denmark Technical University	2024

Conference Presentations (*Talk Given)

- **Narrett, I. S.**, Weiss, B. P., Head, J. W., Lunar Mare Bedrock: Implications for Future Exploration. *Lunar and Planetary Science Conference (LPSC)* (2026)
- Weiss, B. P., Chaffee, T. M., Gattacceca, J., Tikoo, S. M., McDonald, C., Hodges, K. V., Lepaulard, C., Miljković, K., **Narrett, I. S.**, Evidence for a long-lived lunar dynamo. *7th Beijing Earth and Planetary Interior Symposium* (2025)
- ***Narrett, I. S.** et al., Mercury's Ancient Crustal Magnetization: A Stronger Dynamo can be Confirmed by Future BepiColombo Measurements. *EPSC-DPS Joint Meeting* (2025)
- Weiss, B. P., Merayo, J. M. G., Jørgensen, J. L., ***Narrett, I. S.**, Heritage Magnetometers for the Uranus Orbiter and Probe from the Technical University of Denmark. *6th International Workshop on Instrumentation for Planetary Missions* (2025)
- ***Narrett, I. S.** et al., Does Mercury's Ancient Crustal Magnetization Require a Stronger Ancient Dynamo? *LPSC* (2025)
- **Narrett, I. S.** et al., HST and TESS point to a highly spotted photosphere for HD189733A. *TESS Science Conference III* (2024)
- ***Narrett, I. S.** et al., Lunar Crustal Magnetization from Impact-Generated Plasma Amplification of the Lunar Dynamo. *LPSC* and *AGU Fall Meeting* (2024)
- **Narrett, I. S.** et al., Testing Whether a Dynamo Magnetic Field Amplified by Impact Plasmas Can Explain the Magnetization of the Moon. *LPSC* and *ASTRONUM* (both in 2023)
- Strabel, B. P., **Narrett, I. S.**, et al. Magnetic Field Measurement Suite for CubeSat Applications. *AGU Fall Meeting* (2020)